



Laboratory Report

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Student Declaration

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Contents

Description.....	3
Aims.....	3
Method	3
Results and Testing.....	4
Conclusions	4
References	5
Appendices.....	6

Description

This report documents the setup of a working environment between the Windows Host machine and a Linux Virtual Machine. The focus on this lab was to perform the installation of tools such as Python, Visual Studio Code, Git, GitHub Desktop, Notepad++, WinSCP and Putty. The objective was to ensure a communication between both systems using both RDP and SSH, allowing for developmental workflow to prepare for future labs and exercises that involve light programming and scripting.

Aims

1. To Install and configure Python, Visual Studio Code, Git etc on the windows host machine.
2. To clone the GitHub Repository and sync it with local copies of the previous weeks work using the GitHub Desktop application.
3. To install the same list of applications mentioned in step 1 onto the Ubuntu VM.
4. To establish and test a SSH and RDP connection between the Windows host machine and the Ubuntu VM for file transfer and remote access.
5. To prepare both systems for upcoming scripting exercises.

Method

1. Host Machine (Windows 11)

- Installed the latest version of python from the official website ("Python Releases for Windows | Python.org" 2025), ensuring that the **Add Python .exe to PATH** option is selected.
- Installed Visual Studio Code (VSC) from the official website. ("Visual Studio Code - The open source AI code editor" 2025)
- Installed GIT and GitHub Desktop, authenticated both using my credentials and performed the clone option in GitHub Desktop to sync a local copy of my existing repositories. (Week2, Week3). ("Git for Windows" 2025), ("Download GitHub Desktop | GitHub Desktop" 2025)
- Installed Notepad++, PuTTY, WinSCP from official sources, these programs will provide text editing, remote terminal access, and secure file transfer.
- Opened software to ensure all is working as expected and for the likes of Git and Python, opened command prompt and verified installation (e.g., python --version, git --version).

2. Virtual Machine (Ubuntu)

- Installed XRDP using the command `"sudo apt install xrdp -y"` on the Virtual machine so that it can be accessible for access from the host machine by RDP. ("xrdp by neutrino labs" 2025)
- Installed python from the Terminal using the command `"sudo apt install python3 -y"` and to ensure that it was added to PATH.
- Installed and enabled the OpenSSH Server using the command `"sudo apt install openssh-server -y"`. ("OpenSSH server - Ubuntu Server documentation" 2025)
- Installed GIT and GitHub Desktop for Linux following community guides online from GitHub via the terminal used the below commands.

GIT command

- *"sudo apt install git -y"*

GitHub Desktop Commands

- *"sudo apt update && sudo apt upgrade -y"*
- *"wget <https://github.com/shiftkey/desktop/releases/download/release-3.4.1-linux1/GitHubDesktop-linux-amd64-3.4.1-linux1.deb>"*
- *"sudo apt install ./GitHubDesktop-linux-amd64-3.4.1-linux1.deb -y"*

(*"shiftkey/desktop: Fork of GitHub Desktop to support various Linux distributions"* 2025)

- Used GitHub Desktop to sync a local copy of Week2 and Week3 onto the VM.
- Authenticated between both the Linux server and GitHub the *"gh auth login"* command was used.

(*"GitHub CLI | Take GitHub to the command line"* 2025)

- Installed Visual Studio Code from the official Microsoft repository and ensured that it opened and functioned as intended.

Results and Testing

All installations mentioned in the method were successful and both systems operated as intended.

- RDP From Windows host to Linux VM worked successfully using the XRDP Tool. Note: That is uses a slightly different GUI than the Ubuntu GUI Desktop. (See Appendix A)
- Both Windows and Linux scripts to install Python executed correctly. (See Appendix B for Terminal Output's)
- GitHub Desktop successfully synchronised the repos and performed a clone of Week2 and Week3. (See Appendix C)
- Successful authentication between Linux VM Server and GitHub. (See Appendix D)
- Tested that both SSH and WinSCP connections was successful between both Host and VM. (See Appendix E and F)
- Microsoft Visual Studio Installation successful on both Linux and Windows and across both platforms worked as intended.
- No Configuration errors occurred, and successful pings and file transfers tested for a stable network connectivity between host and VM.

Conclusions

The aims of this week's exercise have been achieved successfully. Both the Windows Host machine and the Linux VM had no issues installing of the required applications and tools and connectivity between both machines was verified and tested successfully. This work creates a solid baseline for upcoming exercises that involve cross-platform coding for future labs. Remote access was achieved and demonstrated during this exercise. Understanding and using these applications prepares for more complex exercise to come. Using the likes of SSH and RDP aligns best practices for a more secure way for connecting remotely, SSH provides an encrypted command line option, while RDP provides a more user-friendly GUI (Graphical User Interface) for accessibility. In summary the work carried out in this

exercise strengthens my technical understanding and readiness for advanced automation tasks within the MSc Cloud Technologies Course.

References

Download GitHub Desktop | GitHub Desktop [online] (2025) available:

<https://desktop.github.com/download/> [accessed 10 Nov 2025].

Git for Windows [online] (2025) available: <https://gitforwindows.org/> [accessed 10 Nov 2025].

GitHub CLI | Take GitHub to the Command Line [online] (2025) available:

https://cli.github.com/manual/gh_auth_login [accessed 10 Nov 2025].

OpenSSH Server - Ubuntu Server Documentation [online] (2025) available:

<https://documentation.ubuntu.com/server/how-to/security/openssh-server/index.html>
[accessed 10 Nov 2025].

Python Releases for Windows | Python.Org [online] (2025) available:

<https://www.python.org/downloads/windows/> [accessed 10 Nov 2025].

Shiftkey/Desktop: Fork of GitHub Desktop to Support Various Linux Distributions [online] (2025)

available: <https://github.com/shiftkey/desktop?tab=readme-ov-file#installation-via-package-manager> [accessed 10 Nov 2025].

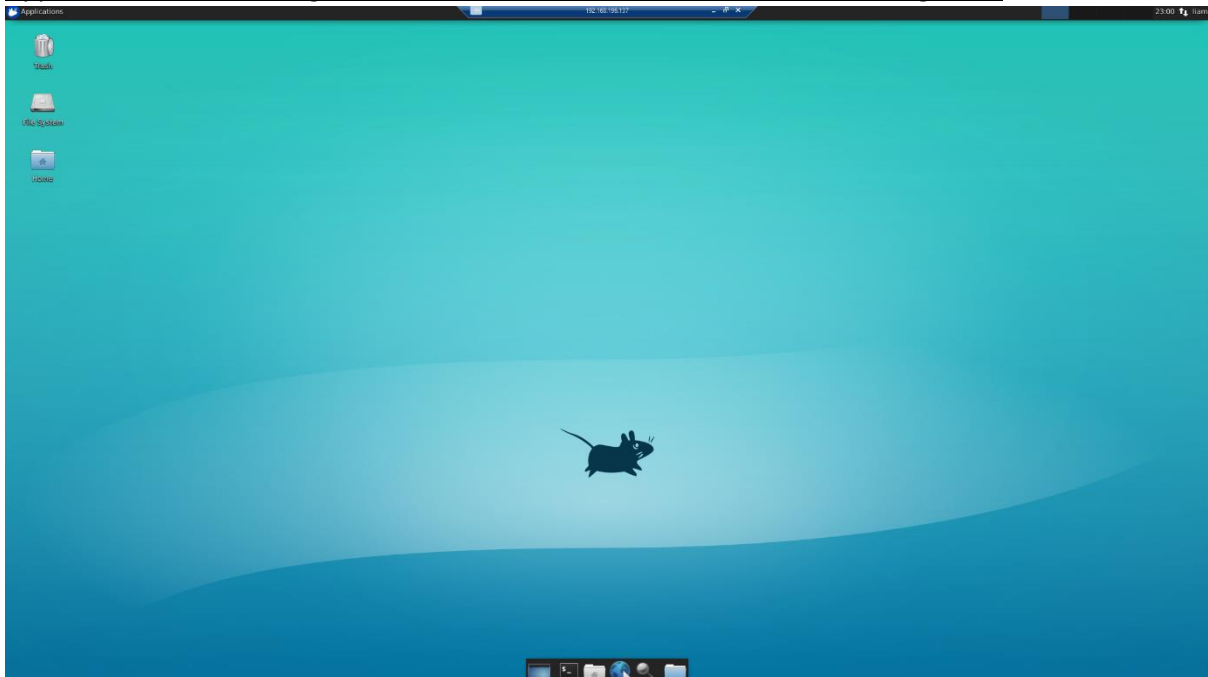
Visual Studio Code - The Open Source AI Code Editor [online] (2025) available:

<https://code.visualstudio.com/> [accessed 10 Nov 2025].

Xrdp by Neutrinolabs [online] (2025) available: <https://www.xrdp.org/> [accessed 10 Nov 2025].

Appendices

Appendix A – RDP working as intended from Windows Host to Linux VM using XRDP



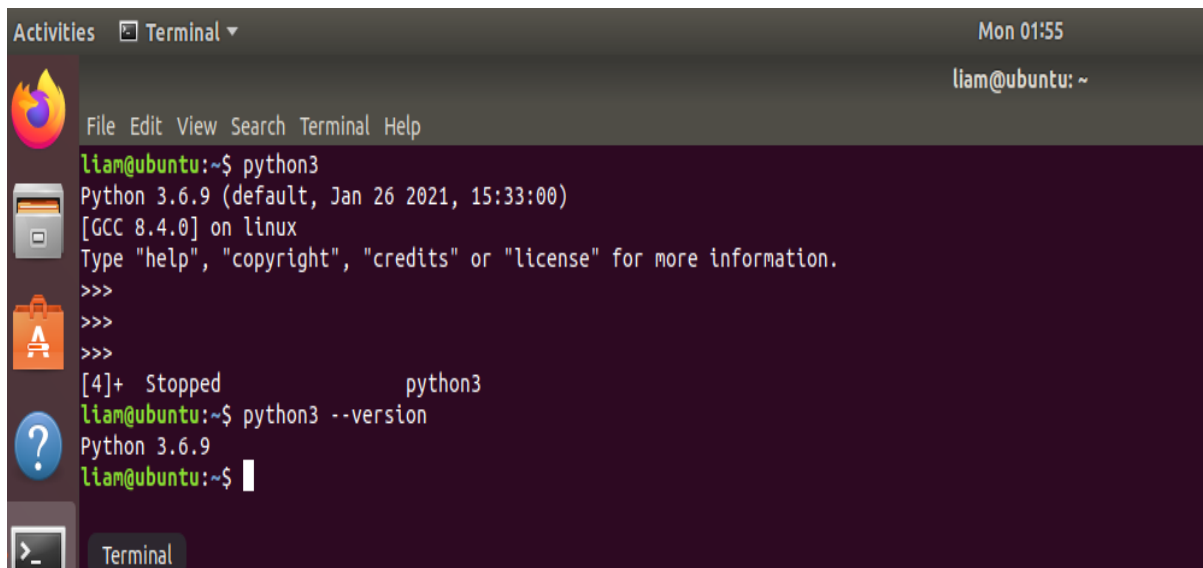
Appendix B - Terminal Output for Python installation on Windows and Linux

```
Command Prompt - python
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Liam>python --version
Python 3.14.0

C:\Users\Liam>python
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> |
```

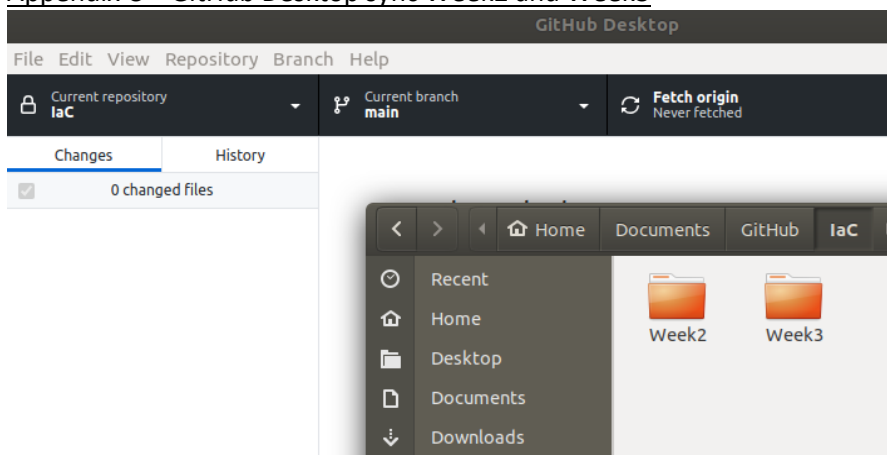
Laboratory Report



A terminal window on an Ubuntu system. The window title is 'Terminal'. The user 'liam@ubuntu' is at the prompt. The user runs 'python3', which outputs 'Python 3.6.9 (default, Jan 26 2021, 15:33:00) [GCC 8.4.0] on linux' and prompts for help. The user then runs 'python3 --version', which outputs 'Python 3.6.9'.

```
liam@ubuntu:~$ python3
Python 3.6.9 (default, Jan 26 2021, 15:33:00)
[GCC 8.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>>
>>>
[4]+ Stopped python3
liam@ubuntu:~$ python3 --version
Python 3.6.9
liam@ubuntu:~$
```

Appendix C – GitHub Desktop sync Week2 and Week3

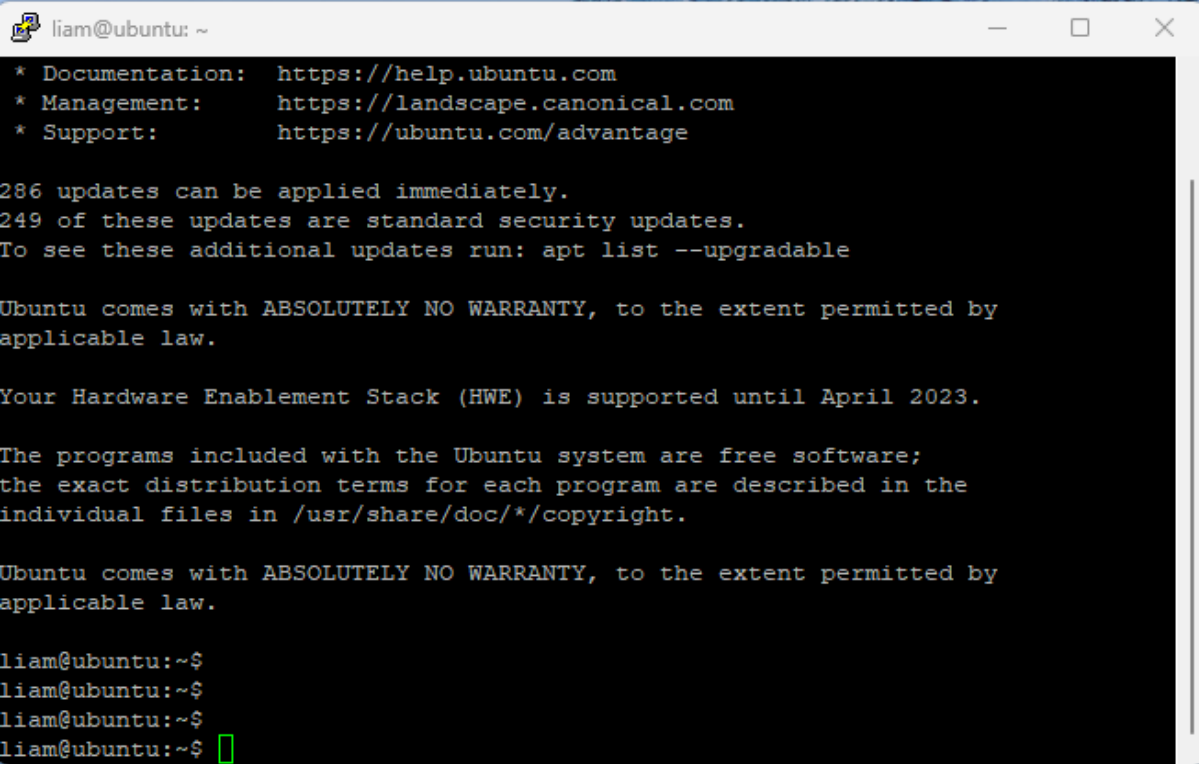


Appendix D – Terminal Output of Authentication for Linux server and GitHub

```
liam@ubuntu:~$ gh --version
gh version 2.83.0 (2025-11-04)
https://github.com/cli/cli/releases/tag/v2.83.0
liam@ubuntu:~$ gh auth login
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? HTTPS
? Authenticate Git with your GitHub credentials? Yes
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: B0E9-7FD7
Press Enter to open https://github.com/login/device in your browser...
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as L00187849
liam@ubuntu:~$
```

Appendix E – Output of Putty terminal using SSH communicating between the Linux VM and Windows Host.



The screenshot shows a Putty terminal window titled 'liam@ubuntu: ~'. The terminal output displays Ubuntu system information, including documentation, management, and support links, update statistics, and warranty information. It also shows the Hardware Enablement Stack (HWE) support duration and program distribution terms. The prompt 'liam@ubuntu:~\$' is repeated four times, with a green cursor visible on the last line.

```
liam@ubuntu: ~
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage

286 updates can be applied immediately.
249 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Your Hardware Enablement Stack (HWE) is supported until April 2023.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

liam@ubuntu:~$
liam@ubuntu:~$
liam@ubuntu:~$
liam@ubuntu:~$
```

Laboratory Report

Appendix F – WinSCP output showing communicating between Linux VM and Windows Host

The image shows the WinSCP interface with two panes. The left pane displays the local Windows file system at 'C:\Users\Liam\Documents\'. The right pane displays the remote Linux VM file system at '/home/liam/'.

Name	Size	Type	Changed
Parent directory		Parent directory	09/11/2025 18:39:17
File folder		File folder	09/11/2025 18:34:12

Name	Size	Changed	Rights	Owner
..		26/10/2025 06:35:35	rwxf-xf-x	root
Desktop		26/10/2025 06:42:17	rwxf-xf-x	liam
Documents		10/11/2025 10:42:25	rwxf-xf-x	liam
Downloads		26/10/2025 06:42:17	rwxf-xf-x	liam
Music		26/10/2025 06:42:17	rwxf-xf-x	liam
Pictures		26/10/2025 06:42:17	rwxf-xf-x	liam
Public		26/10/2025 06:42:17	rwxf-xf-x	liam
snap		10/11/2025 11:03:25	rwxf-xf-x	liam
Templates		26/10/2025 06:42:17	rwxf-xf-x	liam
Videos		26/10/2025 06:42:17	rwxf-xf-x	liam
GitHubDesktop-linux...	95,241 KB	01/07/2024 20:59:22	rw-rw-r--	liam

0 B of 0 B in 0 of 1 5 hidden 0 B of 93.0 MB in 0 of 10 18 hidden

SFTP-3 0:00:24